



SCUD ARSENAL

MAGAZINE



Cybersecurity



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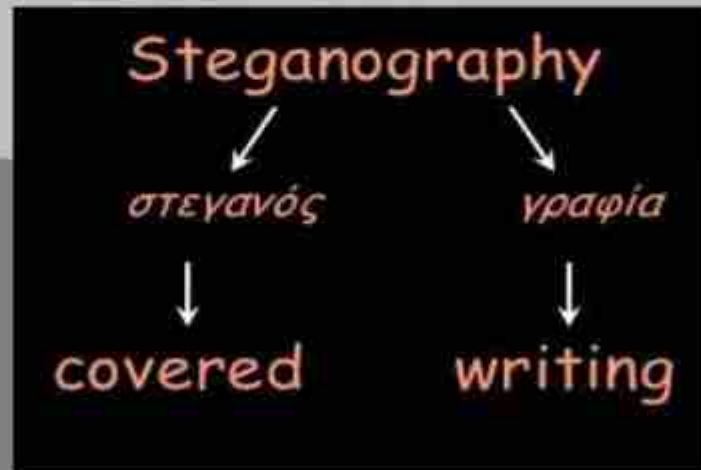
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17A81A0506

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STEGANOGRAPHY



Steganography is the art of hiding information in ways that prevent the detection of hidden messages.

It includes a vast array of secret communications methods that conceal the message's very existence.

In digital steganography, electronic communications may include steganographic coding inside of a transport layer, such as a document file, image file, program or protocol.

Media files are ideal for steganographic transmission because of their large size.

For example, a sender might start with an innocuous image file and adjust the color of every hundredth pixel to correspond to a letter in the alphabet. The change is so subtle that someone who is not specifically looking for it is unlikely to notice the change.

TECHNIQUES

1. PHYSICAL

. Invisible ink

. Tattoo message on head

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- . *Overwrite select characters in printed type in pencil*
- . *Pin punctures*
- . *Microdots*
- . *Newspaper clippings, Knitting instructions, XOXO signatures, report cards*



2. DIGITAL MESSAGES

Modern steganography entered the world in 1985 with the advent of personal computers being applied to classical steganography problems

- . *Concealing messages within the lowest bits of noisy images or sound files.*
- . *Concealing data within encrypted data or within random data.*
- . *Changing the order of elements in a set*
- . *Modifying the echo of a sound file.*

Image of a tree with a steganographically hidden image.

The hidden image is revealed by removing all but the two least significant bits of each color component and a subsequent normalization. The hidden image is shown



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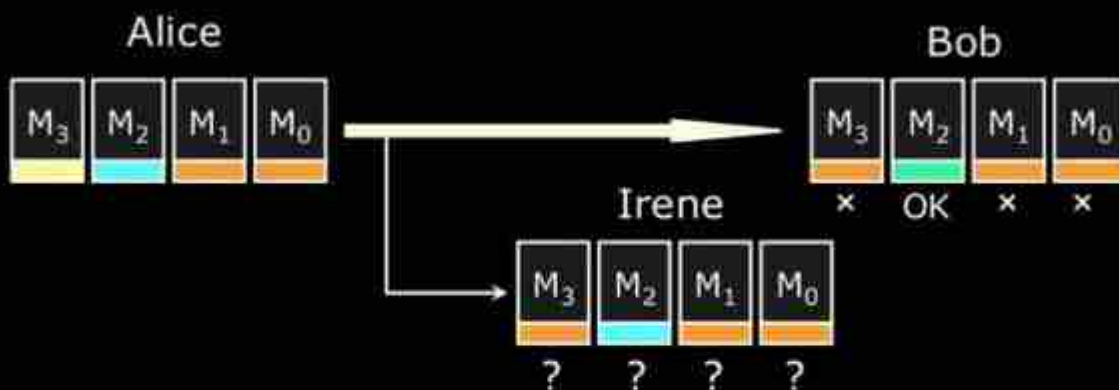
Image of a cat extracted
from the tree image above.



3. CHAFFING AND WINNOWING

Chaffing & Winnowing

- Separate good messages from the bad ones
- Stream of unencoded messages with signatures
 - Some signatures are bogus
 - Need key to test



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4. NULL CIPHER

- .Hide message among irrelevant data
- . Confuse the crypto analyst

Big rumble in New Guinea.
The war on
celebrity acts should end soon.
Over four
big ecstatic elephants replicated.

Bring two cases of beer.

5. NETWORK STEGANOGRAPHY

Contrary to typical steganographic methods that use digital media (images, audio and video files) to hide data, network steganography uses communication protocols' control elements and their intrinsic functionality. As a result, such methods can be harder to detect and eliminate.

6. CYBER PHYSICAL SYSTEMS

Specific techniques hide data in CPS components. For instance, data can be stored in unused registers of IoT/CPS components and in the states of IoT /CPS actuators

7. PRINTED

Some types of modern color laser printers integrate the model, serial number and timestamps on each printout for traceability reasons using a dot-matrix code made of small, yellow dots not recognizable to the naked eye

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auf d auf d
Testo Testo

Yellow dots from a laser printer

Machine ID codes in laser printers



The other techniques include video, audio, UV watermarking, text.

DETECTION

- . Comparing them with previous files
- . All files on suspect file system can be hashed using a hash function and then compared to hash table.
- . A steganography tool can be used to camouflage the secret message in the least

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significant bits but it can introduce a random area that is too perfect. This area of perfect randomization stands out and can be detected by comparing the least significant bits to the next-to-least significant bits on image that hasn't been compressed.

APPLICATIONS

- . Used in modern printers
- . Allegedly used by terrorists
- . Alleged use by intelligence services
- . Digital watermarking
- . Distributed steganography
- . Online challenge

BY

N.V.Sri Lalitha

17A81A0530

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How to Build a Neural Network with Keras

Keras is one of the most popular Deep Learning libraries out there at the moment and made a big contribution to the commoditization of artificial intelligence. It is simple to use and it enables you to build powerful Neural Networks in just a few lines of code. In this post, you will discover how you can build a Neural Network with Keras that predicts the sentiment of user reviews by categorizing them into two categories: positive or negative. This is called Sentiment Analysis and we will do it with the famous imdb review dataset. The model we will build can also be applied to other Machine Learning problems with just a few changes.

Note that we will not go into the details of Keras or Deep Learning. This post is intended to provide you with a blueprint of a Keras Neural Network and to make you familiar with its implementation.

Table of Contents:

- What is Keras?
- What is Sentiment Analysis?
- The imdb Dataset
- Import Dependencies and get the Data
- Exploring the Data
- Data Preparation
- Building and Training the Model

What is Keras?

Keras is an open source python library that enables you to easily build Neural Networks. The library is capable of running on top of TensorFlow, Microsoft Cognitive Toolkit, Theano, and MXNet. Tensorflow and Theano are the most used numerical platforms in Python to build Deep Learning algorithms but they can be quite complex and difficult to use. In comparison, Keras provides an easy and convenient way to build deep learning models. It's creator François Chollet developed it to enable people to build Neural Networks as fast and easy as possible. He laid his focus on extensibility, modularity, minimalism and the support of python. Keras can be used with GPUs and CPUs and it supports both Python 2 and 3. Google Keras made a big contribution to the commoditization of deep learning and artificial intelligence since it has commoditized powerful, modern Deep Learning algorithms that previously were not only inaccessible but also unusable as well.

What is Sentiment Analysis?

With Sentiment Analysis, we want to determine the attitude (e.g the sentiment) of for example a speaker or writer

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with respect to a document, interaction, or event. Therefore it is a natural language processing problem where text needs to be understood, to predict the underlying intent. The sentiment is mostly categorized into positive, negative and neutral categories. With the use of Sentiment Analysis, we want to predict for example a customers opinion and attitude about a product based on a review he wrote about it. Because of that, Sentiment Analysis is widely applied to things like reviews, surveys, documents and much more.

The imdb Dataset

The imdb sentiment classification dataset consists of 50,000 movie reviews from imdb users that are labeled as either positive (1) or negative (0). The reviews are preprocessed and each one is encoded as a sequence of word indexes in the form of integers. The words within the reviews are indexed by their overall frequency within the dataset. For example, the integer "2" encodes the second most frequent word in the data. The 50,000 reviews are split into 25,000 for training and 25,000 for testing. The dataset was created by researchers of the Stanford University and published in a paper in 2011, where they achieved 88.89% accuracy. It was also used within the "Bag of Words Meets Bags of Popcorn" Kaggle competition in 2011.

Import Dependencies and get the Data

We start by importing the required dependencies to preprocess our data and to build our model.

```
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt
import numpy as np
from keras.utils import to_categorical
from keras import models
from keras import layers
```

We continue with downloading the imdb dataset, which is fortunately already built into Keras. Since we don't want to have a 50/50 train test split, we will immediately merge the data into data and targets after downloading, so that we can do an 80/20 split later on.

```
from keras.datasets import imdb

(training_data, training_targets), (testing_data, testing_targets) = imdb.load_data(num_words=10000)

data = np.concatenate((training_data, testing_data), axis=0)
```


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```
targets = np.concatenate((training_targets, testing_targets), axis=0)
```

Exploring the Data

Now we can start exploring the dataset:

```
print("Categories:", np.unique(targets))
```

```
print("Number of unique words:", len(np.unique(np.hstack(data))))
```

Categories: [0 1]

Number of unique words: 9998

```
length = [len(i) for i in data]
```

```
print("Average Review length:", np.mean(length))
```

```
print("Standard Deviation:", round(np.std(length)))
```

Average Review length: 234.75892

Standard Deviation: 173.0

You can see in the output above that the dataset is labeled into two categories, either 0 or 1, which represents the sentiment of the review. The whole dataset contains 9998 unique words and the average review length is 234 words, with a standard deviation of 173 words.

Now we will look at a single training example:

```
print("Label:", targets[0])
```

Label: 1

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```
print(data[0])
```

```
[1, 14, 22, 16, 43, 530, 973, 1622, 1385, 65, 458, 4468, 66, 3941, 4, 173, 36, 256, 5, 25, 100, 43, 838, 112, 50, 670, 2, 9, 35, 480, 284, 5, 150, 4, 172, 112, 167, 2, 336, 385, 39, 4, 172, 4536, 1111, 17, 546, 38, 13, 447, 4, 192, 50, 16, 6, 147, 2025, 19, 14, 22, 4, 1920, 4613, 469, 4, 22, 71, 87, 12, 16, 43, 530, 38, 76, 15, 13, 1247, 4, 22, 17, 515, 17, 12, 16, 626, 18, 2, 5, 62, 386, 12, 8, 316, 8, 106, 5, 4, 2223, 5244, 16, 480, 66, 3785, 33, 4, 130, 12, 16, 38, 619, 5, 25, 124, 51, 36, 135, 48, 25, 1415, 33, 6, 22, 12, 215, 28, 77, 52, 5, 14, 407, 16, 82, 2, 8, 4, 107, 117, 5952, 15, 256, 4, 2, 7, 3766, 5, 723, 36, 71, 43, 530, 476, 26, 400, 317, 46, 7, 4, 2, 1029, 13, 104, 88, 4, 381, 15, 297, 98, 32, 2071, 56, 26, 141, 6, 194, 7486, 18, 4, 226, 22, 21, 134, 476, 26, 480, 5, 144, 30, 5535, 18, 51, 36, 28, 224, 92, 25, 104, 4, 226, 65, 16, 38, 1334, 88, 12, 16, 283, 5, 16, 4472, 113, 103, 32, 15, 16, 5345, 19, 178, 32]
```

Above you see the first review of the dataset which is labeled as positive (1). The code below retrieves the dictionary mapping word indices back into the original words so that we can read them. It replaces every unknown word with a "#". It does this by using the `get_word_index()` function.

```
index = imdb.get_word_index()
reverse_index = dict([(value, key) for (key, value) in index.items()])
decoded = " ".join( [reverse_index.get(i - 3, "#") for i in data[0]] )
print(decoded)
```

this film was just brilliant casting location scenery story direction everyone's really suited the part they played and you could just imagine being there robert # is an amazing actor and now the same being director # father came from the same scottish island as myself so i loved the fact there was a real connection with this film the witty remarks throughout the film were great it was just brilliant so much that i bought the film as soon as it was released for # and would recommend it to everyone to watch and the fly fishing was amazing really cried at the end it was so sad and you know what they say if you cry at a film it must have been good and this definitely was also # to the two little boy's that played the # of norman and paul they were just brilliant children are often left out of the # list i think because the stars that play them all grown up are such a big profile for the whole film but these children are amazing and should be praised for what they have done don't you think the whole story was so lovely because it was true and was someone's life after all that was shared with us all

Data Preparation

Now it is time to prepare our data. We will vectorize every review and fill it with zeros so that it contains exactly

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10,000 numbers. That means we fill every review that is shorter than 10,000 with zeros. We do this because the biggest review is nearly that long and every input for our neural network needs to have the same size. We also transform the targets into floats.

```
def vectorize(sequences, dimension = 10000):  
    results = np.zeros((len(sequences), dimension))  
    for i, sequence in enumerate(sequences):  
        results[i, sequence] = 1  
    return results
```

```
data = vectorize(data)  
targets = np.array(targets).astype("float32")
```

Now we split our data into a training and a testing set. The training set will contain 40,000 reviews and the testing set 10,000.

```
test_x = data[:10000]  
test_y = targets[:10000]  
train_x = data[10000:]  
train_y = targets[10000:]
```

Building and Training the Model

We can now build our simple Neural Network. We start by defining the type of model we want to build. There are two types of models available in Keras: the Sequential model and the Model class used with functional API.

Then we simply add the input-, hidden- and output-layers. Between them, we are using dropout to prevent overfitting. Note that you should always use a dropout rate between 20% and 50%. At every layer, we use "Dense" which means that the units are fully connected. Within the hidden-layers, we use the relu function, because this is always a good start and yields a satisfactory result most of the time. Feel free to experiment with

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other activation functions. And at the output-layer, we use the sigmoid function, which maps the values between 0 and 1. Note that we set the input-shape to 10,000 at the input-layer, because our reviews are 10,000 integers long. The input-layer takes 10,000 as input and outputs it with a shape of 50.

Lastly, we let Keras print a summary of the model we have just built.

```
# Input - Layer
model.add(layers.Dense(50, activation = "relu", input_shape=(10000, )))

# Hidden - Layers
model.add(layers.Dropout(0.3, noise_shape=None, seed=None))
model.add(layers.Dense(50, activation = "relu"))
model.add(layers.Dropout(0.2, noise_shape=None, seed=None))
model.add(layers.Dense(50, activation = "relu"))

# Output- Layer
model.add(layers.Dense(1, activation = "sigmoid"))model.summary()

model.summary()
```

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 50)	500050
dropout_1 (Dropout)	(None, 50)	0
dense_2 (Dense)	(None, 50)	2550

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dropout_2 (Dropout)	(None, 50)	0
---------------------	------------	---

dense_3 (Dense)	(None, 50)	2550
-----------------	------------	------

dense_4 (Dense)	(None, 1)	51
-----------------	-----------	----

=====

Total params: 505,201

Trainable params: 505,201

Non-trainable params: 0

Now we need to compile our model, which is nothing but configuring the model for training. We use the "adam" optimizer. The optimizer is the algorithm that changes the weights and biases during training. We also choose binary-crossentropy as loss (because we deal with binary classification) and accuracy as our evaluation metric.

```
model.compile(  
optimizer = "adam",  
loss = "binary_crossentropy",  
metrics = ["accuracy"]  
)
```

We are now able to train our model. We do this with a batch_size of 500 and only for two epochs because I recognized that the model overfits if we train it longer. The Batch size defines the number of samples that will be propagated through the network and an epoch is an iteration over the entire training data. *In general a larger batch-size results in faster training, but don't always converges fast. A smaller batch-size is slower in training but it can converge faster.* This is definitely problem dependent and you need to try out a few different values. If you start with a problem for the first time, I would you recommend to you to first use a batch-size of 32, which is the standard size.

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```
results = model.fit(  
    train_x, train_y,  
    epochs= 2,  
    batch_size = 500,  
    validation_data = (test_x, test_y)  
)
```

Train on 40000 samples, validate on 10000 samples

Epoch 1/2

40000/40000 [=====] - 5s 129us/step - loss: 0.4051 - acc: 0.8212 - val_loss: 0.2635 -
val_acc: 0.8945

Epoch 2/2

40000/40000 [=====] - 4s 90us/step - loss: 0.2122 - acc: 0.9190 - val_loss: 0.2598 -
val_acc: 0.8950

It is time to evaluate our model:

```
print(np.mean(results.history["val_acc"]))
```

0.894750000536

Awesome! With this simple model, we already beat the accuracy of the 2011 paper that I mentioned in the beginning. Feel free to experiment with the hyperparameters and the number of layers. You can see the code for the whole model below:

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```
import numpy as np

from keras.utils import to_categorical

from keras import models

from keras import layers

from keras.datasets import imdb

(training_data, training_targets), (testing_data, testing_targets) = imdb.load_data(num_words=10000)

data = np.concatenate((training_data, testing_data), axis=0)

targets = np.concatenate((training_targets, testing_targets), axis=0)

def vectorize(sequences, dimension = 10000):

    results = np.zeros((len(sequences), dimension))

    for i, sequence in enumerate(sequences):

        results[i, sequence] = 1

    return results

data = vectorize(data)

targets = np.array(targets).astype("float32")

test_x = data[:10000]

test_y = targets[:10000]

train_x = data[10000:]

train_y = targets[10000:]

model = models.Sequential()

# Input - Layer

model.add(layers.Dense(50, activation = "relu", input_shape=(10000, )))
```

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```
# Hidden - Layers
model.add(layers.Dropout(0.3, noise_shape=None, seed=None))
model.add(layers.Dense(50, activation = "relu"))
model.add(layers.Dropout(0.2, noise_shape=None, seed=None))
model.add(layers.Dense(50, activation = "relu"))

# Output- Layer
model.add(layers.Dense(1, activation = "sigmoid"))
model.summary()

# compiling the model
model.compile(
    optimizer = "adam",
    loss = "binary_crossentropy",
    metrics = ["accuracy"]
)

results = model.fit(
    train_x, train_y,
    epochs= 2,
    batch_size = 500,
    validation_data = (test_x, test_y)
)

print("Test-Accuracy:", np.mean(results.history["val_acc"]))
```



BY

A.N.V.Vamsi Krishna

17A81A0501

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Details of Faculty attended FDPs, Workshops, Seminars, Conference etc., outside the college as well as in the college: 06

S.No	Name and Designation of the Faculty	Name of Workshop/Seminar/ FDP/SDP Attended	Location	No. Of Days	From Date	To Date
1	O .Sri Nagesh	Improving Teaching skills in the subject python programming	JNTUK Kakinada	06	27-05-2019	01-06-2019
2	M.Vamsi krishna	Improving Teaching skills in the subject python programming	JNTUK Kakinada	06	27-05-2019	01-06-2019
3	A leelavathi	Big Data Computing	NIT, AP	06	29-05-2019	04-05-2019
4	A.Rajesh	Big Data Computing	NIT, AP	06	29-05-2019	04-05-2019
5	N.Hiranmayi	Big Data Computing	NIT, AP	06	29-05-2019	04-05-2019
6	B.Sri Ramya	Big Data Computing	NIT, AP	06	29-05-2019	04-05-2019

Important Visitors to the Department: 04

S.No	Name of Eminent Guest	Organization	Date(s) of Visit
1.	Dr.A.Krishna Mohan	Professor, CSE, JNTUK	20.04.2019
2.	Dr. R.B.V. Subramaanyam	Professor, CSE, NITW	20.04.2019
3.	Dr. S. PallamSetty	Professor, CSE, AU	20.04.2019
4.	Mr.SrinivasaRajuVuppalapati	Director, Alykas Innovations Pvt.Ltd	20.04.2019
5.	Mr.EedalaRambabu	Alumni, Member of Technical Staff 2, Micro Focus	20.04.2019

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Student Achievements

a. Internships

S. No	Name	RegdNo	Title	Name Of The Industry	Duration
1	P.Tarun Kumar	15A81A0538	Autocad plugin development	nCircle Tech Pvt.Ltd., A ProtoTech Solutions	11-02-2019 To 31-05-2019
2	Manepalli.Arunadevi	15A81A0529	Embedded Systems	Vaya group, Hyderabad	3-12-2018 To 31-5-2019
3	M.Divya Prasanna	16A81A05K9	Machine Learning with Python	Pivotal Soft Pvt Ltd, Visakhapatnam	4-05-2019 To 4-06-2019
4	S.L Keerthi	16A81A05M5			
5	V.Maha Lakshmi	16A81A05N5			
6	A. Pavani	16A81A05I1			
7	B. VenuDaneswari	16A81A05I4			
8	P.Madhu Annapurna	16A81A05L8			
9	V.C.V.L Mounika	16A81A05N3			
10	V.Navya Srilatha	16A81A05N4			
11	Shaik Sabeen	16A81A05M9			
12	V. Rama Leela Sai	16A81A05N6			
13	N. Devi kataksham	16A81A05L0	IOT	ORL industries	13-05-2019 To 15-06-2019
14	G.Roja Rani	17A85A0502			
15	P. Valli	17A85A0504			
16	P. Sravani	16A81A05L9	Web Design	Kate Technologies, Hyd	16-05-2019 To 18-06-2019
17	K.Krishna Vamsi Sai	16A81A05J6			
18	A.Manikanta	16MU1A0525	Android	Pivotal Soft Pvt Ltd, Visakhapatnam	4-05-2019 To 4-06-2019
19	K.Satya Sri	16A81A05J7	Web Design	Miracle Software Systems	13-05-2019 To 14-06-2019
20	S. Harshini Sai	16A81A05M4	IOT	Mad blocks Technologies Pvt Ltd	13-05-2019 To 13-06-2019
21	K.P.S Sireesha	17A85A0505			
22	T.Karishma Reddy	16A81A05N1	IOT & Robotics		
23	Y. Mounika Lakshmi	16A81A0559			

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24	P. Kanyaka Alekya	17A81A0594	Recruitment Interview Scheduling App	Miracle Software Systems	13-05-2019 To 14-06-2019
25	E. Hema Latha	16A81A05D2	Conference Room Booking and Status App		
26	Amiseti Vamsi Krishna	17A81A0501	Marketing	MYCAPTAIN	25-10-2018 to 31-11-2018

b.The following list of students completed the APSSDC –Gamification workshop , from 16.05.2019 to 31.05.2019.

S.No	Reg. Number	Name of the Student
1	17A81A0501	A N V Vamsi Krishna
2	17A81A0503	BALAGAM SWATHI
3	17A81A0513	GANGULA VENKATA LAKSHMI TULASI
4	17A81A0524	KUCHIBHOTLA SRI SATYA SRAVYA
5	17A81A0543	SIMHADRI TEJA SRI
6	17A81A0557	BALUSU LAVANYA
7	17A81A0566	GODAVARTHI BHARGAVI
8	17A81A0572	JUVVALAPALEPU PRAMEELA
9	17A81A0579	KUNIREDDY DEVI SAI DURGA
10	17A81A0590	PANDIRIPALLI MANJUSHA
11	17A81A0594	PEDAMALLU KANYAKA ALEKYA
12	17A81A05B3	AdabalaSatish
13	17A81A05C4	Chukka Shyam Kumar
14	17A81A05E5	N Teja Kiran
15	17A81A05F4	R Sai Kiran
16	17A81A05F6	B Utthej Kumar
17	17A81A05F9	M V D S Vardhan
18	17A81A05I7	KANCHERLA SAHITHI CHANDRA TEJASWINI
19	17A81A05L4	VEMULA PAVANI SIVA PRATHYUSHA
20	17A81A05L5	VISHNUMOLAKALA BADARINADH
21	17A81A05K4	PULLURI ADILAKSHMI
22	17A81A05H8	CHITTURI LATHA SRAVANI
23	17A81A05H3	BHOGARAJU L N LALITHA PRABHA
24	17A81A05G9	ATYAM N V S S S L RAMYA

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25	17A81A05I5	JAKKI PRAGATHISREE
26	17A81A05H6	CHALLA VENKATA SAI RAM
27	17A81A05I6	GOTETI B S Y VASUDEVA RAO
28	17A81A05H4	BOLLAM RENUKA VENKATA PRIYANKA

c. NPTEL Certified Students List (Jan-April-2019 Session)

S.No.	Roll Number	Name of the student	Course Name	Certificate Type
1	17A81A0595	BhavaniPosimsetty	Programming in Java	Successfully completed
2	17A81A0598	Reddy Akanksha		Elite+Silver
3	17A81A0594	PedamalluKanyakaAleky		Successfully completed
4	17A81A05F4	RayabharapuSaikiran		Elite+Silver
5	17A81A05G7	Sri BhavaniYallavula		Elite+gold
6	17A81A05C5	DamisetiDeepika Naga Ratnam		Elite+gold
7	17A81A05I4	ChaitanyaIndugula	Programming, Data Structures and Algorithms using Python	Elite
8	16A81A05F4	Kommuri Naga SrideviDivya	Data Mining	Elite
9	16A81A0543	SnehithaPenugonda		Successfully completed
10	16A81A0534	MandapakaPavan Kumar		Elite
11	16A81A0539	P Salma Begum		Successfully completed
12	16A81A0533	Surendra Reddy		Successfully completed
13	16A81A0555	SwethaKiranmayi		Successfully completed
14	16A81A0520	ViswaSaiKarpurapu		Successfully completed
15	16A81A05C6	B.V.L.R.Anjali		Elite
16	16A81A05C2	B. Devi Surekha		Elite
17	16A81A05F3	K.Sushmakumari		Successfully completed
18	17A81A0501	Amiseti N V Vamsi Krishna	Introduction to Internet of Things	Elite

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19	18A81D5302	PaletiJyothsnaSree	Big Data Computing	Successfully completed
20	18A81D5301	ChippadaNarayanaBhargav		Successfully completed
21	18A81D5303	PaturiManikantha		Successfully completed

NPTEL Certified Faculty List (Jan-April-2019 Session)

S.N o.	Name of the Faculty	Course	Certificate Type
1.	AnjaniSuputri Devi D	Compiler Design	Elite
2.	Praveen Kumar Nallabariki		Successfully Completed
3.	R.Vinupriya	Computer Architecture And Organisation	Elite
4.	KoradaRajaniKumar		Successfully Completed
5.	LeelavathiArepalli	Problem Solving Through Programming In C	Elite
6.	MademNageswara Rao		Elite
7.	V K Hanuman Turaga	Programming In Java	Elite+Silver
8.	Ch Raja Ramesh		Elite+Silver
9.	R Leele Phani Kumar		Elite+Silver
10.	Shirin Bhanu Koduri		Elite+Gold
11.	Nataraj Gudapaty		Elite
12.	Praneetha Pitta		Elite+Silver
13.	Rama Rajeswari Mulukutla	Programming, Data Structures And Algorithms Using Python	Successfully Completed
14.	VrpsSastryYadavilli		Successfully Completed
15.	LakshminarayanaKondreddi		Elite
16.	SdSabirHussain	Programming In C++	Successfully Completed
17.	A.Sirisha		Successfully Completed
18.	M R Raja Ramesh	Data Base Management System	Elite+Silver
19.	Sukanya		Elite+Silver
20.	ItteLavanya		Elite+Silver
21.	Dr.D Jaya Kumari	Data Mining	Elite
22.	Loshma Guniseti		Elite+Silver
23.	AlamuruVenkannababu		Elite
24.	Sp.Malarvizhi		Elite+Silver
25.	PustuluruHariChandana		Elite
26.	D. SasiRekha	Real Time Operating System	Successfully Completed

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27.	Ch Raja Ramesh	Machine Learning	Successfully Completed
28.	Kavidi Venkata Mutyalu		Successfully Completed
29.	Kavidi Venkata Mutyalu	Social Networks	Successfully Completed
30.	Gokavarapu Sriram Ganesh	Introduction To Automata, Languages And Computation	Elite+Silver

Sayonara-2k19 conducted on 04.04.2019

Placements

Roll No.	Name of the Student	Company	CTC (LPA)
15A81A0501	ALLA LAKSHMI BHARGAVI	TCS CODE VITA	3.6 LPA
15A81A0528	MAJETY H V A SRI VYSHNAVI		
15A81A0536	NEELAM PRUDHVINI SAI		
15A81A0546	THOTA NEERAJA		
15A81A05C0	VUDDAGIRI INDUMATHI		
15A81A05F1	MADDAMSETTI LAKSHMI MOUNICA		
15A81A0514	KEDARISETTI DEEPIKA	MPHASIS	1.8 LPA
15A81A0517	KALLA SATYAVATHI		
15A81A0536	NEELAM PRUDHVINI SAI		
15A81A0546	THOTA NEERAJA		
15A81A0562	AINAMPUDI LAVANYA DEVI		
15A81A0576	KANDUKURI AKHILA		
15A81A0588	MAMIDALA PURNA VAMSI		
15A81A0595	DURGAPRIYA NOWBATHULA		
15A81A0597	PAILU KAVYA PRIYA		
15A81A05B4	SUREDDY KRISHNA SAI PHANI KUMAR		
15A81A05B9	VANDANAPU LASYAPRIYA		
15A81A05C4	ALLAM SNEHA SRI SANDHYA		
15A81A05F2	VARALAKSHMI MAHANATHI		
15A81A05G8	SARIDE KARTHIK		
15A81A05H0	SEELAM SOWJANYA		
15A81A0515	KALAVALAPALLI BHASKAR GANESH	L&T TECHNOLOGY	4.0 LPA

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15A81A0552	VARANASI LAKSHMIMEGHANA		
15A81A0555	VELIGATLA SHAMITHA		
15A81A0565	ATMAKURI SAI RUCHITHA		
15A81A0573	GONTLA LAKSHMI INDIRA		
15A81A05A2	PENDYALA DURGA SUNEETHA		
15A81A05A6	SAMAYAMANTHULA J LAKSHMI SRI		
15A81A05C0	VUDDAGIRI INDUMATHI		
15A81A05D1	CHILUKURI LAKSHMI CHANDINI		
15A81A05D9	GRANDHI PRAGATHI RAO		
15A81A05F1	MADDAMSETTI LAKSHMI MOUNICA		
15A81A05F5	MUSUNURU BALA KRISHNA		
15A81A05J5	DONE RESHMA DHAR		
15A81A05K0	KARRI CHIRANJEEVI SAI ISWARYA		
15A81A05L3	MANGENA GIRI NAGA PAVAN KUMAR		
15A81A05M6	POSINA SASANKA		
15A81A05M8	SAJJA SAI KUMAR		
16A85A0503	KAVALA SAI RAMESH		
15A81A0510	GARLAPATI NAVYA	TCS NINJA	3.5 LPA
15A81A0547	TUMMALAPALLI SUPRIYA		
15A81A0554	V.VINEELA		
15A81A0555	VELIGATLA SHAMITHA		
15A81A0565	ATMAKURI SAI RUCHITHA		
15A81A0595	DURGAPRIYA NOWBATHULA		
15A81A0596	PADMANABHUNI MANIDEEP		
15A81A0597	PAILU KAVYA PRIYA		
15A81A05A1	PATCHIPULUSU A L SOWMYA SRI		
15A81A05D7	G.HEMALATHA		
15A81A05G2	YAMUNA PANCHADI		
15A81A05I0	Y.BHANUPRIYA		
15A81A05K9	KUSAM VENKATA ALEKHYA		
15A81A05M0	NARRA MOUNIKA		
15A81A0525	KOTHA NALINI VENKATA LAKSHMI	EFFTRONICS SYSTEMS PVT. LTD.,	3 - 7 LPA
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15A81A0580	KODE SUBHASHINI		
15A81A05F8	NEKKANTI VYSHNAVI		
15A81A05K1	KATTA VASAVI VIDYA SIREESHA		
15A81A0586	MADUPALLI JAYANTH	SNOVASYS SOFTWARE SOLUTIONS PVT. LTD.,	3.4 LPA
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15A81A05A6	JHANSI L S SAMAYAMANTHULA		
15A81A05B1	SRIPADA DIVYA SATYA SAI		
15A81A05C3	ALLA NAGAVALLI DEVI		
15A81A05C9	BUDDANA SRAVYA TULASI		
15A81A05F8	NEKKANTI VYSHNAVI		
15A81A05G8	SARIDE KARTHIK		
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15A81A05K0	KARRI CHIRANJEEVI SAI ISWARYA		
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15A81A05G2	PANCHADI YAMUNA		
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16A85A0505	KOSURI SUVARNA RAJU		

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SNIPPETS

WHAT IS THE OUTPUT OF FOLLOWING :

1.

```
#include<stdio.h>

int main()
{
    signed char chr=128;
    printf("%d\n",chr);
    return 0;
}
```

a)128
b)-128
c)based on compiler
d)None
2.

```
#include<stdio.h>

int main()
{
    short i;
    for(i=1;i>=0;i++)
    printf("%d\n",chr);
}
```

a. control won't fall into loop
b. numbers will be displayed until signed limit and throw runtime error
c. numbers will be displayed until signed limit and program terminates
d. it will be infinite loop printing numbers without error

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3. `#include <stdio.h>`

`void main()`

`{`

`int k = 0;`

`for (k)`

`printf("Hello");`

`}`

a) Compile time error

b) hello

c) Nothing

d) Varies

4. `#include <stdio.h>`

`void main()`

`{`

`int k = 0;`

`for (k < 3; k++)`

`printf("Hello");`

`}`

a) Compile time error

b) Hello is printed thrice

c) Nothing

d) Varies

5. `#include <stdio.h>`

`void main()`

`{`

`double k = 0;`

`for (k = 0.0; k < 3.0; k++)`

`printf("Hello");`

`}`

a) Run time error

b) Hello is printed thrice

c) Hello is printed twice

d) Hello is printed infinitely

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6. `#include <stdio.h>`

`int main()`

`{`

`int a = 1, b = 1, c;`

`c = a++ + b;`

`printf("%d, %d", a, b);`

`}`

a) a = 1, b = 1

b) a = 2, b = 1

c) a = 1, b = 2

d) a = 2, b = 2

7. `#include <stdio.h>`

`int main()`

`{`

`int a = 10, b = 10;`

`if (a = 5)`

`b--;`

`printf("%d, %d", a, b--);`

`}`

a) a = 10, b = 9

b) a = 10, b = 8

c) a = 5, b = 9

d) a = 5, b = 8

8. `#include <stdio.h>`

`int main()`

`{`

`int i = 0;`

`int j = i++ + i;`

`printf("%d\n", j);`

`}`

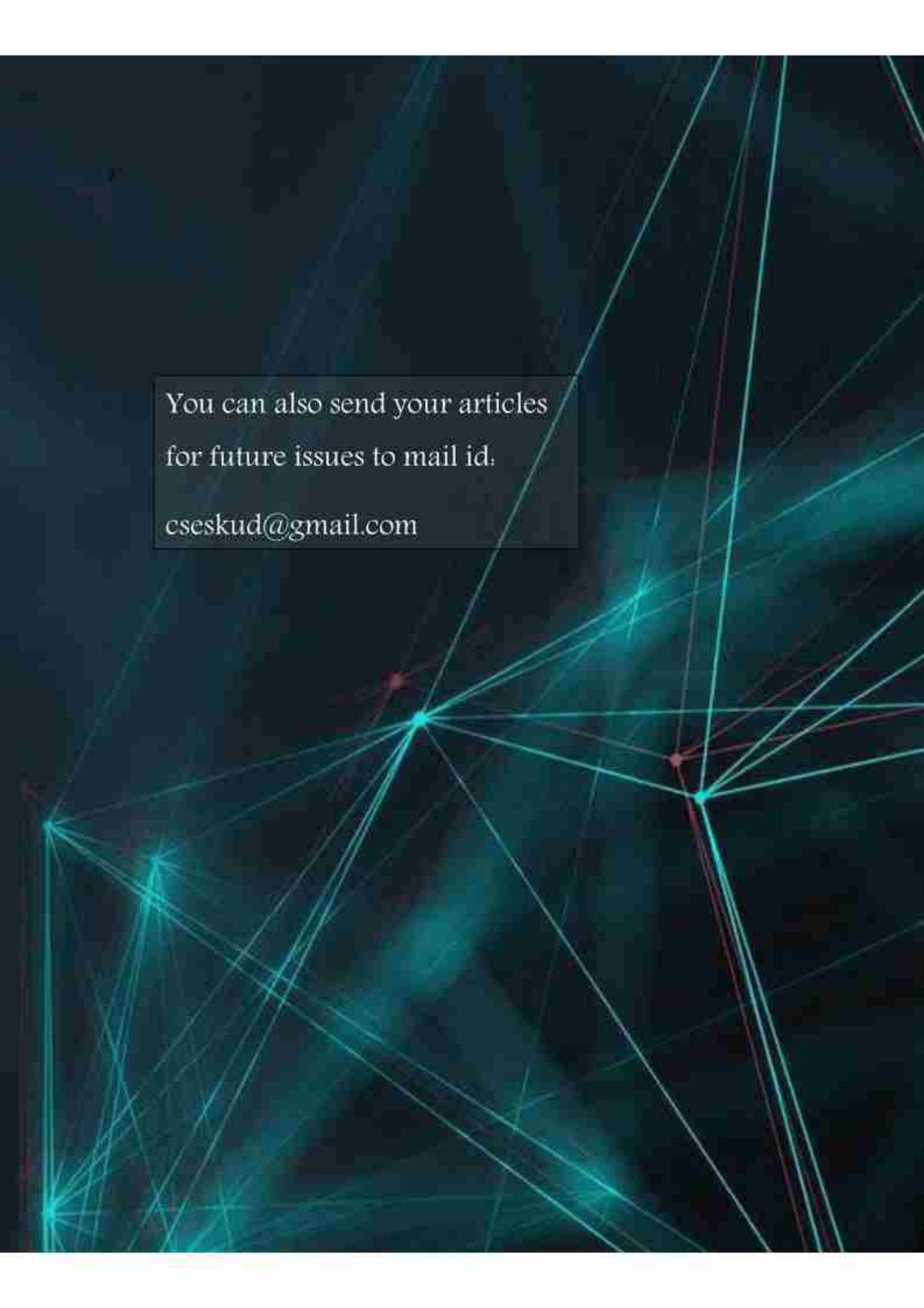
a) 0

b) 1

c) 2

d) Compile time error

ANSWERS: 1-b , 2-c , 3-a , 4-a , 5-b , 6-b , 7-c , 8-b



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